CBB Originator Interconnected Player Prop Predictive Model

Data, Modeling, Simulation & Future In-Play Architecture — Blueprint v1.3

Purpose

Create a disciplined, no-drift plan for building a pregame college-basketball originator that produces a coherent joint distribution for every player on the court, while preserving a clean architectural path to a separate in-play model later.

Core decision
Build the pregame model first. Treat the in-play model as a separate predictive model that shares the same basketball state representation, player/team latent talent estimates, lineup representations, event logic, and simulation framework. Do not allow in-play implementation to delay the pregame build.

Primary output board

MIN	PTS	TS%	REB	AST	STL	BLK	TOV	PF	ORB	DRB	FT	FT%	2P	2P%	3P	3P%
per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player	per player

Internally, the model must also represent FTA, FTM, 2PA, 2PM, 3PA, 3PM, FGA and FGM. Percentages and composite statistics are derived from simulated primitive events rather than independently predicted.

Document status

Item	Decision
Pregame model	Build first; production priority.
In-play model	Separate future model; architect for it now, implement later.
Simulation	Exact PMF/CDF when tractable; otherwise convergence-qualified joint simulation. Interactive low-N simulation is diagnostic only.
Data acquisition	Phase 0 and hard gate before predictive-model coding.
Sportsbook benchmark	Required only to demonstrate edge; benchmark data is separate from the pure-originator prediction path.
Project governance	Frozen data/model contracts + explicit acceptance gates to prevent drift.

Prepared: October 4, 2026

1. Executive decisions

The strongest architecture is a hierarchical, dynamic, lineup-aware joint generative model whose posterior predictive distribution is propagated through a possession-level simulator. The model should simulate the basketball game and allow player props and game markets to emerge from that common probability universe.

Do not build 17 independent prop models.
Independent models can produce good marginal projections while violating basketball accounting identities and misrepresenting teammate/opponent dependence. The central object is the joint game distribution P(Y | X), not a collection of unrelated marginal forecasts.

Seven non-negotiable principles

- Pregame and in-play are separate predictive models, with a shared modeling platform.
- Data correctness and point-in-time integrity outrank model sophistication.
- Availability, minutes and rotation are modeled explicitly; they are not afterthoughts.
- Feature importance is learned statistically. The simulator propagates probabilities and enforces mechanics; it does not decide what is predictive.
- Interdependence should arise primarily from shared basketball events, lineups, finite opportunities and latent game factors—not from a correlation matrix pasted onto independent marginals.
- Every sophisticated candidate must beat simpler challengers in untouched walk-forward testing.
- Claims of sportsbook superiority require no-vig market comparisons, calibration, CLV and realized out-of-sample evidence on benchmark samples; historical sportsbook data is not required to train the pure basketball model.

System-level architecture

```
PREGAME INFORMATION
      ↓
LATENT PLAYER / TEAM TALENT
      ↓
AVAILABILITY → ROTATION / MINUTES → FIVE-MAN LINEUPS
      ↓
GAME ENVIRONMENT + MATCHUP + PLAYER/LINEUP INTERACTIONS
      ↓
POSSESSION / EVENT PROBABILITIES
      ↓
AUTHORITATIVE JOINT DISTRIBUTION ENGINE
(EXACT WHEN TRACTABLE; CONVERGED MC OTHERWISE)
      ↓
COHERENT PLAYER BOARD + GAME MARKETS + JOINT PROBABILITIES
      ↓
CALIBRATION / PRICING / SPORTSBOOK BENCHMARKING
```

2. Two predictive models — explicitly separate

2.1 Pregame Originator model

The pregame model predicts the full joint distribution before tipoff using only basketball information that existed before the chosen prediction timestamp. Its information set includes player talent, projected availability, expected minutes and lineups, opponent strength, pace, matchup, venue/rest/travel context and other basketball-derived features. Sportsbook lines and player props are excluded from the flagship prediction path.

Canonical target:

```
P(Y_game | information available before tipoff)
```

2.2 In-Play Originator model

The in-play model begins after the game starts and predicts the remainder of the game conditional on actual observed state. It is separately trained, separately validated and separately deployed because its statistical problem is different: it must combine strong pregame priors with streaming evidence and state transitions.

Canonical target:

```
P(Y_remaining | pregame posterior, current score/clock, current lineups, live events, live availability)
```

2.3 What the models share

Shared component	Why share it
Canonical IDs and schemas	Avoid duplicate mapping logic and inconsistent player/team identity.
Latent player/team talent	Pregame estimates become priors for live updating.
Lineup representation	The same five-man interaction structure matters before and during games.
Event definitions/accounting rules	A shot, assist, rebound, turnover, etc. must mean the same thing in both systems.
Simulation engine mechanics	Pregame simulates the full game; in-play simulates only the remaining game.
Validation utilities	Calibration, scoring rules and market comparison should use consistent definitions.

Important boundary
The in-play model must not be built now. We only make architectural choices that preserve its future path. The pregame originator must be finished and validated first.

3. Pregame model — formal architecture

3.1 Model the primitives, derive the box score

The displayed board contains statistics that are algebraically related. Predicting all of them independently creates incoherence. The internal model therefore simulates primitive events and derives the public board.

Displayed statistic	Internal treatment
MIN	Direct stochastic rotation/lineup outcome.
PTS	Derived: 2×2PM + 3×3PM + FTM.
REB	Derived: ORB + DRB.
TS%	Derived from points, FGA and FTA using the chosen canonical

Displayed statistic	Internal treatment
	definition.
FT%, 2P%, 3P%	Derived from makes/attempts; undefined when denominator is zero under an explicit convention.
AST/STL/BLK/TOV/PF	Generated from possession/event processes.
Team points / total / margin	Aggregated directly from simulated player/game events.

3.2 Layered probability model

- **Layer 1 — Point-in-time data**: Only information available before prediction time enters training or inference.
- **Layer 2 — Dynamic latent talent**: Hierarchically shrunk, time-evolving player/team estimates of relatively stable skills.
- **Layer 3 — Availability**: Active/out/uncertain/minutes-restriction state with timestamped evidence.
- **Layer 4 — Rotation and minutes**: Probabilistic minutes and five-man units constrained to team-minute accounting.
- **Layer 5 — Player/lineup interaction**: Hierarchical main effects plus low-rank teammate/opponent interaction representations.
- **Layer 6 — Game environment**: Shared latent pace, shooting, foul, turnover and rebounding environment.
- **Layer 7 — Possession/event models**: Conditional probabilities for turnovers, shots, shooters, shot type/result, fouls, FTs, assists, blocks, rebounds and steals.
- **Layer 8 — Joint numerical distribution engine: use exact probability summation wherever the fitted state admits a tractable exact PMF; otherwise generate complete possible games with a converged joint simulator. Never use a low-draw interactive simulation as the pricing source.**
- **Layer 9 — Deterministic accounting**: Build the player board and game markets from the simulated events.
- **Layer 10 — Calibration**: Correct systematic probability/distribution errors without breaking accounting.
- **Layer 11 — Joint-dependence validation**: Test whether the model captures co-movement, conditional probabilities and tail dependence.
- **Layer 12 — Pricing: convert the authoritative distribution into P(over), P(under), P(push) and fair prices. Exact PMF/CDF sums are authoritative when available; otherwise use a convergence-qualified simulation bank with recorded Monte Carlo uncertainty.**

3.3 Separate talent, role and environment

State family	Examples	Expected update speed
Talent	Shooting, passing, rebounding, turnover avoidance, steal/block ability.	Slow
Role	Minutes, starter probability, usage, ball-handling share, shot share, rebound opportunity.	Medium / fast
Environment	Pace, opponent lineup, matchup, venue, foul/turnover/rebounding conditions.	Game-specific

This separation is essential for both pregame accuracy and the later live model. A one-game shooting spike should not be treated as an instant permanent change in talent, while a starting-role change or teammate absence can legitimately move role expectations immediately.

3.4 Rotation and lineup model

Minutes are not independent regressions. Regulation college basketball imposes approximately 200 team player-minutes before overtime, and players compete for finite court time. The model should first create an uncertain rotation state and then generate feasible five-man lineups/stints.

```
Availability → starter probabilities → player minute shares → feasible five-man units → stint durations
```

The advanced version can use a semi-Markov substitution process conditioned on clock, score, foul state, current lineup, opponent and coach tendencies. In the two-week v1, a constrained stochastic rotation model plus historical substitution/lineup priors is the more efficient build.

3.5 Possession/event engine

Each possession is conditioned on the ten players on the floor, the game environment and current context. The fitted models estimate conditional event probabilities; the simulator samples them.

```
Possession
  → turnover OR foul OR field-goal attempt
  → if FGA: shooter → shot family/location → make/miss/block
  → if make: possible assist
  → if miss: rebound contest → ORB/DRB → possible continuation
  → if foul: foul type → FT sequence / possession consequence
  → next state / substitution / possession
```

3.6 Interdependence

Relationship	How dependence is generated
PG AST <-> teammate PTS	Same made basket can credit both events.
Shooter miss <-> rebounder REB	Same missed shot creates the rebound opportunity.
STL <-> opponent TOV	Same turnover event.
BLK <-> opponent FGA outcome	Same contested shot event.
Teammate minutes	Finite 200-minute team constraint.
Teammate usage / shot volume	Finite possessions and shot opportunities.
Opponent players	Shared pace, score state, lineups and matchup environment.
All count stats	Shared number of possessions and game duration.

3.7 Residual dependence layer

Basketball mechanics will not explain every remaining covariance. After fitting the structural model, calculate residuals and test whether significant dependence remains. Only then add a low-dimensional residual latent-factor layer or a carefully justified copula. The copula is a residual correction, not the central architecture.

4. Why simulation is mathematically advantageous

Monte Carlo does not create predictive signal. It is a numerical integration tool used when the fitted joint distribution cannot be enumerated exactly. If an exact PMF/CDF is available for a market, direct probability sums are authoritative and simulation is only a diagnostic. If the full CBB generative distribution is not analytically enumerable, a sufficiently

converged simulation bank becomes the numerical representation of that fitted distribution, subject to explicit Monte Carlo error and convergence gates.

4.1 Probability authority, pricing, and simulation policy

The public product must never present two apparently competing answers for the same probability without explaining the numerical relationship. The authoritative probability source depends on whether the fitted distribution can be enumerated exactly.

Case	Authoritative pricing source	Role of simulation	Public presentation
Exact PMF/CDF is available	Direct sums of the exact distribution.	Diagnostic/visualization only; never overrides exact pricing.	Show one exact probability. Do not place a noisy low-N estimate beside it as an equal answer.
Full joint distribution is not tractably enumerable	Convergence-qualified Monte Carlo representation of the fitted generative model.	Primary numerical integration mechanism.	Show the model probability from the production draw bank, plus a quality/convergence flag when appropriate.
Interactive user simulation	Never a pricing source.	Educational/diagnostic sample from the model.	Label as Simulation Laboratory and disclose sampling noise and simulation count.

Custom-line pricing must use the same authoritative distribution as the displayed model probabilities. For an exact discrete distribution, calculate spread, total, team-total and other line probabilities by direct summation of the PMF/CDF. For a simulation-only distribution, query the full production simulation bank rather than running a new small interactive sample.

Exact-versus-simulation comparisons belong in an internal validation view. When an exact benchmark exists, use a large fixed-seed diagnostic run (normally at least 100,000 draws and higher when needed), report Monte Carlo standard error or confidence intervals, and test distribution-level discrepancies such as maximum CDF error and total-variation distance. Individual low-probability PMF bins must not be judged by raw relative percentage differences from a small sample.

Public UX rule: separate Exact/Production Pricing from the Simulation Laboratory. Plain-language line semantics (for example, “-3.5 covers with a win by 4+”) should be primary; formulas belong in methodology/help text. A low-N simulation is never allowed to create a second public probability that appears to challenge the authoritative model probability.

```
For simulation s = 1,...,S:
   Y^(s) ~ P(Y | X, fitted parameters)

P(PTS_A > 20.5) ≈ count[PTS_A^(s) > 20.5] / S
P(PTS_A > 20.5 AND AST_B > 6.5) ≈ count[both conditions] / S
```

Simulator rule
The simulator must never be the place where “feature importance” is manually decided. Feature engineering, estimation, shrinkage, ablation and proper scoring determine predictive relationships. The simulator simply propagates them through possible games.

4.2 Coherence gates

- PTS_i = 2×2PM_i + 3×3PM_i + FTM_i.
- REB_i = ORB_i + DRB_i.
- Team player-minutes reconcile to regulation/overtime minutes under explicit accounting.
- Team points equal the sum of simulated player scoring plus any explicitly modeled official team scoring adjustments.
- An assist requires an eligible made field goal under the chosen stat rules.
- A steal must reconcile to an opponent turnover event.
- Percentages are computed from simulated numerators and denominators.
- Every simulation stores a deterministic seed/version/data-cutoff/roster hash for reproducibility.

4.3 Injury and DNP recalculation

```
NEW STATUS: Player A = OUT
      ↓
Remove Player A from feasible states
      ↓
Re-estimate starter / minutes / rotation distribution
      ↓
Re-estimate five-man lineup frequencies
      ↓
Reallocate ball handling, usage, shot mix, assist and rebound opportunities
      ↓
Recalculate opponent interactions and game environment
      ↓
Resimulate the entire joint game distribution
      ↓
Publish new full board + game markets + correlations
```

This is conditional inference, not retraining. The fitted model is reused with a new information state, making last-minute recomputation feasible.

5. Blueprint changes intended to maximize predictive accuracy

5.1 Model opportunity before efficiency

For player props, opportunity errors can dominate. The build should devote exceptional attention to availability, starter probability, minutes, rotation, possessions, role and attempts before obsessing over fine-grained conversion models.

5.2 Hierarchical shrinkage for CBB sparsity

College basketball has freshmen, transfers, rapid role changes, uneven schedules and many small samples. Player, team and lineup parameters should partially pool toward position/class/conference/team-level priors. Exact five-man units should not receive unrestricted parameters unless supported by sufficient data.

5.3 Low-rank player/lineup interaction representation

Represent teammate and opponent interactions with shrunk low-dimensional embeddings/factorization terms. This allows the model to learn meaningful pair and lineup effects without requiring large samples for every exact five-man combination.

5.4 Attempts/opportunities separated from conversion

Stat family	Opportunity process	Conversion/outcome process
Scoring	Minutes → possessions → usage → FGA/FTA	2P/3P/FT make probabilities

Stat family	Opportunity process	Conversion/outcome process
	→ shot type	
Assists	Ball-handling/creation opportunities + teammate shot opportunities	Assist credited on eligible made field goals
Rebounds	Missed-shot opportunities + on-floor position/role	Rebounder allocation; ORB vs DRB
Steals/blocks	Opponent possession/shot opportunities	Event-specific defensive probabilities

5.5 Challenger models are mandatory

Model	Purpose
Baseline A — rolling/rate distributions	Simple benchmark; catches overengineering.
Baseline B — independent CatBoost/LightGBM heads	Strong marginal benchmark.
Baseline C — independent hierarchical statistical models	Strong probabilistic/shrinkage benchmark.
Candidate D — full joint generative model	Must justify its complexity with superior out-of-sample performance.

The joint model is not protected from competition. If a simpler challenger wins on untouched tests for a target or subsystem, the simpler method should be used there.

5.6 Market isolation and benchmark layer

PURE ORIGINATOR: P(Y | basketball information only)

BENCHMARK LAYER (read-only): sportsbook game lines / player props are used only for comparison and CLV/edge analysis, never as flagship predictive inputs.

The flagship CBB Originator is intentionally independent of Vegas: P(Y | basketball information only). Sportsbook spreads, totals and player props are maintained in a separate read-only benchmark layer. A market-aware challenger may be researched later, but it is outside the core build and must never contaminate the pure-originator training, model selection or published probabilities.

6. Future in-play model — planned now, built later

The in-play model should be a separate sequential state-space model with a remaining-game simulator. Pregame posterior distributions become live priors; live events update them. The observed part of the game is fixed and only the remainder is simulated.

```
Pregame posterior
    + live event stream
    ↓
Sequential Bayesian / state-space update
    ↓
Current posterior over pace, role, lineup, efficiency, etc.
    ↓
Simulate only the remaining game
    ↓
Final player/game distributions = observed statistics + remaining distribution
```

6.1 Do not apply a single arbitrary prior-decay schedule

State	How quickly live evidence should move it
Starting lineup / current rotation	Very fast
Minutes trajectory / foul trouble / availability	Very fast
Pace / possession environment	Fast
Usage / shot-volume role / assist opportunity	Medium-fast
Rebound/turnover environment	Medium
Underlying 3P/FT/finishing talent	Slow

Bayesian updating should determine the weight of new evidence. An unexpected starting lineup is highly informative immediately; two missed threes should barely move a strong prior on true shooting ability.

6.2 Candidate live inference method

A Rao-Blackwellized particle filter / Sequential Monte Carlo framework is a natural long-term candidate because the state mixes continuous quantities (pace, efficiency, usage) and discrete quantities (lineup, possession, substitution/availability states) and is not naturally linear-Gaussian. This is a future research choice, not a Day-1 implementation requirement.

6.3 Live data contract

Live field	Requirement
Clock / period / score	Mandatory
Every PBP event + event timestamp	Mandatory
Players on floor / substitutions	Mandatory
Player/team foul and bonus state	Mandatory
Current player box score / minutes	Mandatory
Possession/team with ball	Strongly preferred
Timeouts / ejections / injury events	Preferred / mandatory where relevant
Corrections/deletions to prior events	Mandatory
Live sportsbook prices + timestamps	P1 for market benchmarking/trading evaluation; not an input to the flagship live-originator state update.
Provider event time / received time / processed time / prediction time	Mandatory for latency-correct backtesting

7. Phase 0 — data acquisition before predictive-model coding

Hard gate
No core predictive-model coding begins until the Data Readiness Gate passes. Data engineering can proceed in parallel with provider procurement, but no modeling component may quietly invent unavailable fields or silently substitute a

different information set.

7.1 Recommended source roles

Role	Recommended source	Reason
Historical basketball backbone	CollegeBasketballData (CBBD)	P0 source for games, player/team stats, PBP, on-floor players, substitutions, lineups, shooting, rosters, recruiting/transfers and other basketball context.
Current pregame availability	Sharp Stats	P0 production source for current-season injury/absence reason, status, update date and note. Our receipt timestamp creates PIT history going forward.
Independent QA + market benchmark	BALLDONTLIE NCAAB (BDL)	P1 reconciliation of player/team/game outcomes and preferred prospective opening/current odds benchmark with both-side prices and timestamps.
Historical player-strength enrichment	Bart Torvik	P1 non-blocking enrichment for explicit DNPs, game-level advanced player rates, shot-location splits and BPM/OPM/DPM. Core build proceeds without it if acquisition is unsuitable.
Future low-latency live PBP	Sportradar or comparable enterprise feed	Future in-play project only; must not delay Pregame v1.

7.2 What the attached Starter Pack already provides

The attached schema shows player-game minutes, pace, usage, points, rebounds, assists, steals, blocks, turnovers, fouls, starter status and shooting makes/attempts; player-season shooting profiles; and team-game possessions, pace, efficiency and Four Factors. The attached notebooks also show 21 seasons of team/player summary coverage in several modules and 12 seasons of player shooting-profile coverage in the example environment. Those assets are valuable for feature engineering and priors, but the uploaded conversation does not include the underlying raw CSV files themselves.

7.3 Mandatory pregame datasets

Dataset	Minimum fields	Priority
Game metadata	IDs; exact tip; venue; home/away/neutral; tournament; OT; conference	P0
Player game logs	MIN; start; PTS; FGA/M; 2PA/M; 3PA/M; FTA/M; ORB/DRB; AST/STL/BLK/TOV/PF	P0
Team game logs	Possessions; pace; ORtg/DRtg; shooting; rebounds; TO; FT; Four Factors	P0
Play-by-play	Event type; clock; score; actors; shot	P0

Dataset	Minimum fields	Priority
	result/type/location	
Players on floor	Five offensive + five defensive players by event/stint	P0
Substitutions	Player in/out; clock; period; score state	P0
Five-man lineup history	Players; seconds; possessions; pace; ratings; event totals	P0
Rosters / player identity	Persistent player ID; team; class; position; size where available	P0
Availability / injury history	Status; effective/published/received timestamps; restriction if known; 2021-22 to 2025-26 preferred	P0 for automated availability layer; core performance engine can proceed without older history
Historical player props	Book; player; market; line; both-side prices; timestamp if collected	P1 benchmark only; NOT a model input and NOT build-blocking
Historical game markets	CBBD provider; spread; total; ML; explicit opening spread/total; verify stored-line closing semantics	P1 benchmark/context; dense timestamped movement is P2
Recruiting / transfers	Rating/rank/stars; prior team; destination; eligibility	P0/P1
Schedule / rest	Prior game date/location; rest; travel context	P0
Venue metadata	Location; elevation if feasible	P1

7.4 Injury data — required for automated availability, not the underlying performance engine

Historical injury labels are not foundational to the core model. The historical player-performance and absence-response layers are trained from actual participation/DNP, starts, minutes, substitutions, lineups, and teammate redistribution. Current pregame availability is still P0 production information and is supplied prospectively by Sharp Stats.

```
player_id
game_id
status
status_effective_at
provider_timestamp
received_at
injury_type
minutes_restriction_if_known
source
```

Do not delay the build attempting to reconstruct a five-year injury archive. Begin append-only Sharp Stats capture now, preserve the raw structured status and descriptive note, and record our own exact received_at timestamp. Multi-year historical injury/status archives are P2 enrichment only.

7.5 Historical absence-response training table

```
absent_player
replacement_starter
minutes_reallocation
usage_reallocation
```

```
assist_share_reallocation
shot_mix_reallocation
lineup_frequency_change
pace_change
team_efficiency_change
opponent_efficiency_change
rebounding_share_change
```

This table is used to learn conditional role/rotation responses rather than implementing proportional hand-written injury adjustments.

Historical DNP/participation history is P0. For each scheduled player-game candidate, preserve played/DNP, starter state, minutes, prior role, lineup/substitution participation and teammate responses. CBBD is primary; Bart Torvik explicit DNP and advanced game logs are optional P1 enrichment/verification.

7.6 Secondary QA and enrichment sources

BALLDONTLIE NCAAB (BDL) is a secondary source for independent player/team/game reconciliation and prospective sportsbook benchmarking. Its odds feed includes both-side spread/total prices plus timestamps, making it preferable to dense Odds API backfills for the initial benchmark layer. Bart Torvik is a non-blocking enrichment source for explicit DNPs, advanced player-game rates, shot-location buckets and BPM/OPM/DPM. Neither source replaces CBBD's PBP/on-floor/substitution/lineup backbone.

7.7 Lineup stints

```
game_id | start_clock | end_clock
home_1..home_5 | away_1..away_5
possessions | score_start | score_end
```

CBBD lineup aggregates are useful, but reconstructed stints from PBP + on-floor information + substitutions are even more valuable because they preserve the sequence and game state needed for rotation and event models.

7.8 Event-level shot table

```
shooter | shot_family | location | range | made
assisted | assister | players_on_floor | score_state | clock
offensive_rebound_sequence
```

This supports the critical split between opportunity/attempt prediction and conversion efficiency.

7.9 Cold starts: freshmen and transfers

The model must have explicit priors for players with little current-season evidence. Recruiting rating/rank, position, height/weight, class, prior team/conference and prior performance should inform initial latent talent and role distributions. As current-season evidence accumulates, the posterior naturally moves away from the prior.

8. Point-in-time integrity and bitemporal storage

The most important data-engineering rule is that historical features must be reconstructable exactly as they existed at the forecast timestamp.

```
X_(game,t) = information actually available before prediction time t
```

Every externally sourced observation should retain multiple timestamps rather than only the event date.

```
event_time
provider_timestamp
received_timestamp
effective_timestamp
ingested_timestamp
```

8.1 Why this matters

If a player is ruled out at 6:43 PM for a 7:00 PM game, a T-60 prediction must not know it while a T-10 prediction may know it. A database that stores only the final status silently introduces lookahead bias and can make a mediocre model appear excellent.

8.2 Source lineage

Every canonical record should preserve source, source ID, provider update timestamp, retrieval timestamp and schema version. Raw vendor payloads remain immutable.

9. Canonical data lake and schemas

```
data/
├── raw/
│   ├── cbdd/
│   ├── injuries/
│   ├── odds/
│   └── live/
├── canonical/
│   ├── games.parquet
│   ├── teams.parquet
│   ├── players.parquet
│   ├── rosters.parquet
│   ├── player_games.parquet
│   ├── team_games.parquet
│   ├── plays.parquet
│   ├── substitutions.parquet
│   ├── lineup_stints.parquet
│   ├── availability_snapshots.parquet
│   ├── recruiting.parquet
│   ├── transfers.parquet
│   ├── game_odds.parquet
│   └── player_prop_odds.parquet
└── features/
    ├── pregame/
    └── live/
```

Raw files are immutable. Canonical tables are deterministic transformations with versioned schemas. Feature tables are explicitly prediction-time aware.

9.1 Different subsystems may use different history windows

Subsystem	Potential history window logic
Long-run team/player priors	Use the deepest reliable history available.
Shot-profile model	Use seasons where shot-family tracking is reliable.
Lineup interaction model	Use only seasons with adequate on-floor/lineup coverage.
Absence-response model	Use the deepest reliable

Subsystem	Potential history window logic
	participation/DNP/minutes/rotation history from CBBD; multi-year injury diagnoses are not required. Sharp Stats availability history begins prospectively.
Sportsbook benchmark	Use BDL current/opening odds prospectively at aligned checkpoints; CBBD lines are secondary context. Odds API is fallback only. Historical player props are not required.
Bart Torvik enrichment	Use seasons with stable permitted acquisition and reliable player/date mapping; non-blocking and tested only as an incremental feature/QA layer.

This avoids throwing away useful long-run information simply because a newer, richer dataset has a shorter history.

10. Data Readiness Gate

Before the model build officially begins, produce measured—not assumed—coverage statistics by season and data family.

Gate	Acceptance condition / required audit
Player IDs	Target ≥99.5% stable historical identity mapping; exceptions documented.
Team IDs	100% canonical mapping for modeled games.
Game IDs	100% canonical mapping across joined sources.
Player/team box scores	Coverage quantified by season/team; missingness mechanisms documented.
PBP	Coverage quantified by season/team.
Players on floor	Coverage quantified; usable-event percentage reported.
Substitutions	Coverage quantified and cross-checked against minutes/lineups.
Lineups	Coverage quantified; exact seasons eligible for lineup model identified.
Shot locations/families	Coverage and coding consistency measured.
Historical availability	Timestamped/replayable history quantified for the automated availability layer; lack of older history triggers a documented fallback, not a block on the performance engine.
Player-prop benchmark history	Optional historical benchmark coverage documented if collected; NOT a core Data Readiness gate.
Game-line benchmark history	CBBD opening/stored-line coverage verified and stored-line closing semantics tested. Dense timestamped movement is optional; prospective snapshots are preferred.
Future live PBP	Observed latency and correction behavior tested.
Point-in-time replay	Historical feature snapshot reproducible for arbitrary game/checkpoint.

Gate	Acceptance condition / required audit
Licensing	Intended use permitted by every production source.
Historical DNP/participation	Coverage and roster/game candidate construction measured; unexplained missing rows distinguished from true DNP.
Sharp Stats mapping	Current-season player/team mapping rate, status/note conflicts, snapshot freshness and receipt timestamps measured.
BDL reconciliation	CBBD-vs-BDL player/team/game outcome disagreement rate reported; no silent overwrite.
Bart Torvik enrichment	If enabled, player/date/team mapping and field coverage measured; failures do not block core v1.

Coverage rule
Never write “the source has lineups/PBP/injuries” and move on. Quantify the usable percentage by season. The eligible training window for each subsystem is determined by actual coverage.

11. Validation and sportsbook-superiority test

11.1 Marginal accuracy

- MAE/RMSE for point estimates where useful.
- Log score / negative log likelihood.
- CRPS / quantile loss for distributions.
- Brier score and log loss for held-out threshold events; sportsbook prop lines are needed only when evaluating the model at a sportsbook-specific offered line.
- Calibration curves and expected calibration error.
- Prediction interval coverage at multiple levels.

11.2 Joint-distribution accuracy

- Energy score for multivariate forecasts.
- Variogram score / targeted dependence scoring.
- Predicted vs realized teammate/opponent correlations by stat family.
- Conditional probabilities and co-hit rates for logically related props.
- Tail/joint calibration, not just mean correlation.

11.3 Market benchmark

```
BENCHMARK LAYER ONLY

sportsbook no-vig probability  <->  pure-originator probability

Compare where contemporaneous prices exist by:
- Brier / log loss
- calibration
- CLV / line-direction agreement
- realized ROI after actual vig
- market / book / edge bucket / time-to-tip
```

Historical player-prop and dense line-movement backfills are optional. Prospective 2026-27 snapshots are preferred. Sportsbook data never enters the flagship prediction path.

A lower RMSE on player points is not sufficient evidence that the model is better than a sportsbook. Compare against contemporaneous sportsbook probabilities only where a valid benchmark snapshot exists. Historical player props are optional for this purpose; prospective 2026-27 snapshots can establish the benchmark cleanly.

11.4 Injury-specific backtests

Create historical event studies for actual absences, surprise starts, minutes restrictions and multiple-player absences. Where timestamped injury history exists, also evaluate what was knowable pregame. The reallocation model should beat simple proportional or role-average rules without requiring historical sportsbook data.

12. No-drift project governance

To protect the two-week build from architecture drift, freeze five machine-auditable contracts before modeling.

Contract	Purpose
DATA_REQUIREMENTS.md	Every required field, priority and reason.
SOURCE_MATRIX.md	Exact provider/endpoint for each field; availability/cost/license notes.
POINT_IN_TIME_POLICY.md	Exactly what a forecast at time t may use.
CANONICAL_SCHEMA.md	Canonical IDs, column definitions, units and null conventions.
COVERAGE_AUDIT.json	Machine-readable coverage and pass/fail results by source/season.
PROBABILITY_ENGINE_POLICY.md	Defines distribution authority, exact-PMF versus Monte Carlo rules, custom-line pricing, simulation convergence/MCSE gates, and public versus diagnostic presentation.

12.1 Change-control rule

Once the contracts pass review, coding agents may not silently redesign schemas, substitute data, broaden model scope or add new providers. Any proposed change must be recorded as an explicit blueprint change with rationale, expected benefit, cost and validation test.

12.2 Block rather than improvise

```
BLOCKED_DATA_REQUIREMENT
- missing field/source
- affected subsystem
- exact evidence
- proposed options
- no hidden workaround
```

Source freeze: no further data-source shopping or architecture redesign before implementation unless a P0 access/coverage gate fails. Historical injury archives are explicitly non-blocking.

13. Time-boxed build plan

The two-week model clock should begin only after required provider credentials/data access are available. Procurement and source verification can run in parallel before that clock starts.

When	Workstream	Deliverable	Gate
Pre-build	Data procurement	Freeze CBBD + Sharp Stats + BDL + non-blocking Bart Torvik source architecture; confirm credentials/access; no historical injury archive required.	Access confirmed
Day 1	Raw ingestion + IDs	Build canonical IDs/raw-store scaffold; ingest CBBD core identity/game/player/team data; add BDL ID maps/QA scaffold.	Join audit passes
Day 2	Coverage + PIT gate	Ingest CBBD PBP/substitutions/lineups; integrate Sharp Stats current-season availability snapshots; begin coverage and reconciliation audit.	Data Readiness Gate
Day 3	Canonical tables/features	Canonical DuckDB/Parquet + PIT feature scaffolding; historical DNP/participation construction; optional Torvik adapter if non-blocking.	Leakage tests pass
Day 4	Baselines	Rolling + independent ML + hierarchical challenger baselines.	Benchmark suite
Day 5	Availability/rotation	Availability/rotation/minutes model using historical participation/DNP/rotation plus current Sharp Stats conditioning; 200-minute coherence.	Minutes calibration gate
Day 6	Lineup/stints	Reconstruct stints; lineup priors; player interaction representation.	Stint reconciliation
Day 7	Latent talent/game environment	Dynamic shrunk skill + pace/environment states.	Walk-forward likelihood
Day 8	Possession event heads	TO/foul/shot + actor selection.	Event calibration
Day 9	Shot/assist/rebound chain	Shot type/result, assist, block, rebound, FT sequences.	Accounting tests
Day 10	Joint simulator	Full coherent games + board/game-market aggregation.	Simulation invariants
Day 11	Injury recalculation	Conditional re-rotation/reallocation +	Scratch stress tests

When	Workstream	Deliverable	Gate
		fast resimulation.	
Day 12	Calibration	Marginal + distributional calibration; residual dependence test.	Calibration gate
Day 13	Full walk-forward benchmark	Compare Candidate D vs all basketball challengers on full walk-forward tests; compare to sportsbook only on valid historical/prospective benchmark samples.	Model-selection gate
Day 14	Production hardening	Speed, reproducibility, manifests, monitoring, final acceptance.	Release candidate

13.1 What does not belong in the two-week critical path

- Building the in-play model.
- Optical player-tracking dependencies.
- Social-media sentiment ingestion.
- Complex orchestration infrastructure before the basic jobs need it.
- A giant end-to-end transformer simply because it is sophisticated.
- Manual feature proliferation without ablation evidence.

14. Efficient implementation stack

Layer	Recommended v1 choice
Language	Python 3.12
Storage	Parquet + immutable raw payloads
Local analytical engine	DuckDB
Dataframes / columnar	Polars / PyArrow
Statistical/hierarchical modeling	PyMC/Stan or efficient custom GLM/Bayesian components where justified; do not force one framework
Tree baselines	LightGBM / CatBoost
Interaction embeddings	PyTorch only where low-rank interactions justify it
Simulation	Exact PMF/CDF when tractable; otherwise convergence-qualified joint simulation. Interactive low-N simulation is diagnostic only.
Serving	FastAPI or equivalent lightweight service
Artifacts	Versioned model manifest + data cutoff + code SHA + seed + roster hash
Scheduling	Simple reproducible jobs first; avoid heavyweight orchestration until needed

15. Licensing and commercial-use constraint

The attached Starter Pack license states that the bundle is licensed for personal, non-commercial use by the original purchaser and prohibits incorporating the bundle into a commercial or paid product without written permission. It also prohibits redistribution or public hosting of the bundle. Therefore, private research/prototyping can use the Starter Pack under its terms, but any commercial deployment must rely on appropriately licensed production data or obtain the required permission.

Production rule
Do not let the prototype accidentally hard-depend on Starter Pack files if the intended deployment would be commercial. Keep provider adapters and canonical schemas separate so the production data source can be swapped without rewriting the modeling system.

16. Release acceptance criteria for Pregame v1

- Data Readiness Gate passed and documented.
- No point-in-time leakage found in automated replay tests.
- Minutes/rotation distributions calibrated and team-minute coherent.
- All hard accounting identities pass for every generative Monte Carlo draw; exact-distribution paths pass the equivalent deterministic accounting checks.
- Late injury/DNP scenarios recompute the full board without retraining; historical injury data improves the automated availability layer but is not required for the basketball-event engine itself.
- Candidate model compared against all challenger baselines in walk-forward testing.
- Marginal probabilities are calibrated by stat family using basketball outcomes; sportsbook prices are reserved for a separate benchmark layer.
- Joint-dependence metrics show improvement over independent baselines.
- Game-level totals/margins reconcile exactly with the authoritative distribution source: direct PMF aggregation where exact, or player-event simulations where the joint distribution is simulation-based.
- Pure-originator results are primary. Any future market-aware challenger is reported separately and cannot affect the pure model.
- Sportsbook comparison uses contemporaneous no-vig probabilities where available; historical player-prop and dense line-movement backfills are not prerequisites for model construction.
- Production run stores reproducibility metadata and completes inside the agreed latency budget. Every published probability records whether it came from `exact_pmf` or `converged_mc`; simulation-based probabilities also record draw count and Monte Carlo uncertainty.

Where an exact distribution exists, fixed-seed high-N simulation diagnostics must fall within prespecified Monte Carlo tolerances; failures block release and trigger sampling/indexing/support-reconciliation review.

Interactive Simulation Laboratory draws are never persisted as the official price source, and public UI must not present low-N simulated PMFs as peer estimates beside exact probabilities.

If pricing depends on Monte Carlo because the full joint distribution is not exactly enumerable, adaptive or sufficiently large simulation must meet a frozen convergence/MCSE gate before publication.

If an exact PMF/CDF exists for a published market, the public/custom-line probability equals the direct distribution sum; a finite simulation estimate cannot replace it.

17. Evidence and source map

The blueprint combines the attached Starter Pack materials with current public documentation for external data sources. External provider capabilities must still be verified against the exact paid tier/contract before procurement.

Source	Use in blueprint
Attached: headers(1).md	Starter Pack CSV schemas: games, player-game/season stats, player

Source	Use in blueprint
	shooting, team-game/season stats.
Attached: 00_data_dictionary.ipynb	Starter Pack data overview and example historical coverage.
Attached: 06_opponent_adjusted_ratings.ipynb	Opponent-adjusted/SRS concepts and example season coverage.
Attached: 09_player_rate_stats.ipynb	Per-40/per-possession player-rate concepts and example season coverage.
Attached: 10_player_shot_profile_analysis.ipynb	Shot-family and assisted-shot profile concepts; example shooting-history coverage.
Attached: 12_matchup_explorer.ipynb	Tempo-free matchup, recent-form and Four Factors concepts.
Attached: LICENSE	Starter Pack personal/non-commercial license and redistribution restrictions.
CBBD Lineups API	https://api.collegebasketballdata.com/api/lineups
CBBD Plays/Substitutions API	https://api.collegebasketballdata.com/api/plays
CBBD Lines API	https://api.collegebasketballdata.com/api/lines
CBBD Recruiting/Transfer API	https://api.collegebasketballdata.com/api/recruiting
CBBD Teams API	https://api.collegebasketballdata.com/api/teams
CBBD Games/Scoreboard API	https://api.collegebasketballdata.com/api/games
SportsDataIO NCAA Basketball Workflow Guide	https://sportsdata.io/developers/workflow-guide/ncaa-basketball
SportsDataIO NCAA Basketball API Documentation	https://sportsdata.io/developers/api-documentation/ncaa-basketball
Sportradar NCAAM Play-by-Play	https://developer.sportradar.com/basketball/reference/ncaamb-play-by-play
User-provided BDL NCAAB OpenAPI	Secondary player/team/game QA; current/opening odds with both-side prices and timestamps; active-player QA.
User-provided Sharp Stats injury examples/glossary	Current-season availability source: status, reason, update date, note; requires our own exact snapshot receipt time.
Bart Torvik historical player game logs (user-provided example)	P1 non-blocking enrichment for explicit DNPs, advanced rates, shot-location buckets and BPM/OPM/DPM.
DATA_SOURCE_POLICY_v1_0.md	Frozen source responsibilities and market-isolation rules.
MODEL_ARCHITECTURE_FREEZE_v1_0.md	Frozen architecture and change-control conditions.

Research background informing the architecture

The architecture is also consistent with established concepts from possession-level basketball modeling, hierarchical player/lineup estimation, state-space/sequential Bayesian updating, and multivariate proper scoring. Those research threads should be converted into a formal model-specification bibliography when the mathematical specification is frozen; the immediate engineering priority remains data readiness and reproducible point-in-time backtesting.

18. Final blueprint position

The data-source and historical-availability architecture is now frozen: CBBD is the P0 basketball backbone; Sharp Stats supplies current availability; BDL supplies independent QA and prospective market benchmarks; Bart Torvik is optional non-blocking player-strength enrichment. Historical injury archives are not a prerequisite. Implementation should begin without further source-shopping unless a P0 coverage gate fails.

Pregame mission
Build one probabilistic basketball game—not a pile of independent player projections. Every player distribution, game market and joint probability should be a coherent consequence of the same conditioned basketball state.

The pregame system is the immediate product. The separate in-play model is the next product and should inherit the pregame posterior, shared state definitions and simulation mechanics while adding its own sequential updating, live rotation logic, latency-aware data pipeline and remaining-game validation. The source architecture and SOURCE_MATRIX are now frozen. The next implementation stage is CANONICAL_SCHEMA.md + POINT_IN_TIME_POLICY.md, immediately followed by Data Platform v1.0 ingestion and the measured COVERAGE_AUDIT.json.